Upstream Supply Chain Optimization for The Home Depot

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Abstract

The Home Depot's logistics, focusing on the upstream supply chain, consisting of vendors, trucks, personnel, and distribution centers, would greatly benefit from an in-depth study of their data and processes to streamline operations and reduce the time it takes for goods to reach consumers. The Home Depot has access to vast amounts of data from all aspects of its supply chain, and applying data science techniques to this data could provide critical insights for optimization. The goals of the analysis are to study and understand the current bottlenecks and areas where issues arise in the supply and delivery chain, and then suggest ways to improve the entire process. This study could be conducted on multiple levels, from ordering from vendors to selecting more efficient delivery routes to stores and customers, and streamlining distribution center operations. Measures of success include reducing costs for delivery, increasing delivery timeliness and accuracy, and reducing the time required for distribution center movements. All these measures would lead to happier consumers who would receive their products faster and have more of the goods they need or desire in stock at stores. Happier consumers would lead to increased profits for The Home Depot. The Home Depot stock would invariably rise as well, generating additional profits for shareholders and the company.

Problem Description

In 2007, Home Depot had a very decentralized direct-to-store distribution model. Each store handled its own ordering and inventory management, consuming as much as 60% of store labor (Bond, 2015). This drove The Home Depot to build Rapid Deployment Centers (RDCs). Suppliers and vendors ship goods to the RDCs, which then disseminate them throughout the rest

of the company's supply chain. "Upon arrival at the RDC, product might be cross-docked and allocated to stores based on real-time demand (Bond, 2015)." In 2018, The Home Depot intended to invest approximately $1.2 billion in its supply chain. It made "big strides in its upstream supply chain… but feels there is still some way to go to fully leverage its upstream network, as well as build the fastest and most efficient delivery system" (Team Trefis, 2018). According to a Supply Chain video on The Home Depot corporate site, the RDCs are "equipped with advanced automation and technologies that sort, label, and transport product from receiving to shipping, leveraging this technology to minimize handling and enable fast and efficient delivery to our stores" (The Home Depot, 2018).

Current operations at The Home Depot's warehouses and distribution centers can become chaotic. One of the most significant issues is the docks. There are multiple types of docks, including some that are automated, others that use only robots for unloading, and some that employ people for this task. What is in the trailer is the deciding factor on which dock it needs to go to. Sometimes shipments are sent ahead of time or not tracked well, and the warehouse or distribution center only finds out about the shipment when the trailer arrives. This leads to a scramble to determine which dock door the trailer should be directed to for unloading. Once in the warehouse, there are again multiple options as goods are unloaded and moved around. Nothing stays in one place in the warehouse, and it is unloaded on one side of the center and then moved to the other. There are several processes for moving items around the distribution center. One process involves using the PandA (Print and Apply) System to tag merchandise that meets the size requirements for the robot system. Alternatively, merchandise that exceeds these requirements must be tagged by a person, especially for goods that are too large for a conveyor

belt. On the other side of the distribution center are areas designated for each store's inventory needs. Therefore, every product entering the warehouse needs to be separated and tagged for the specific store it will be traveling to (Anonymous, personal interview, 15 September 2025).

Business Understanding

While numerous aspects of The Home Depot's logistics could be improved, the focus of this paper will center on upstream supply chain management and the Rapid Distribution Center (RDC). "An RDC is not a warehouse; it is a large flow-through facility that typically serves about a hundred stores. RDCs do last-minute allocations and deploy to the stores as quickly as possible, resulting in better in-stock positions and higher service levels" (Armstrong, Charles, 2011). The question being answered here is how data science models can optimize the RDC process, from a trailer arriving at a store to departing for another store. This process involves receiving vendor shipments at the distribution center, moving them through the center, and packing trailers to distribute products to stores. The goal is to utilize data science to provide feedback on current processes, thereby improving the performance of the upstream supply chain. Desired outcomes include decreasing the costs of gas needed for trailer trucks, requiring fewer workers or increasing worker productivity in the distribution centers, and reducing the time products spend moving around the distribution center before reaching their destination.

The Home Depot is continually seeking ways to shorten the timeline of its upstream supply chain. "In terms of its upstream network, Home Depot has made a meaningful improvement in devising an optimal flow network with its RDCs (Rapid Deployment Centers), improving its inventory management capabilities, and driving further efficiency through the

Supply Chain Sync initiative (Team Trefis, 2018)" A more streamlined process would increase profits for the company, making the whole executive team happy. Their compensation is mainly equity-based and aligns with company performance and stock (Simply Wall Street, 2025). The Home Depot stock price would increase, making shareholders pleased. The Home Depot also offers profit sharing to select staff, meaning workers benefit from increased revenue.

Data Understanding

The data required to utilize data science for optimizing suggestions in the upstream supply chain, focusing on the RDC, spans multiple avenues and systems. The Home Depot has a myriad of data from all facets of the upstream supply chain, starting with the trucks traveling to the RDCs. "Home Depot also uses Freight Logistics Optimization Works (FLOW), which was initiated by the U.S. Department of Transportation as a data-sharing platform to ease cargo flow during the COVID-19 pandemic (Ross, 2025)." According to the U.S. Department of Transportation, FLOW provides access to a broad, daily view of the current conditions of the overall logistics network, beyond what a single business can observe within its own operations. This data is available only to companies that provide qualified data, and it cannot be shared publicly. This data includes regional supply data, purchase orders, incoming container volumes, and transportation modes. All of this enables any company participating in the FLOW program to review demand fluctuations and mitigate bottlenecks and service-level volatility.

Back in 2015, The Home Depot recognized that to remain competitive in the market, it needed to deliver faster and at lower costs than its competitors. This led them to start digitizing their supply chain around 2015. This began by connecting stores to their website and building the

RDC. Realizing that their entire supply chain could be further optimized, "Home Depot introduced Project Sync, a software designed to improve metrics from the beginning to end of the supply chain, allowing trucks, workers, and machines to operate as efficiently as possible" (HBS 2019). Sync provides data on supplier/carrier information, capacity constraints, and pricing to enable the development of optimized instructions and schedules. The stock for every truck, materials that need to be unloaded, and diagrams of how to position carts and pallets are all available through Project Sync. At a 2023 Investor Conference, John Deaton, Executive Vice President of Supply Chain & Product Development, discussed The Home Depot's investment in automating and mechanizing the Rapid Deployment Center to reduce the number of product touches and expedite the movement of goods.

This automation is evident in the databases for the RDC, which contain a large amount of data on the facility, including incoming and outgoing trailers, as well as workers. There is data available for the RDC itself regarding the number of trailers in a given yard. The dock door data type determines the types of goods that can be unloaded at that dock. This data details the location of the trailers, the status of the trailer (whether it is at the distribution center or not), and whether it is being loaded, unloaded, or simply waiting. There is also data on the number of people needed to unload the truck, as well as the number of forklifts required if necessary. The trailer data also includes the RDC yard it is in, the number of different SKUs in that trailer, and whether it has been queued to go to a door. The number of cartons on the trailer and the estimated time to unload it are also available. As teammates unload the trailer, they scan a box as it is brought off the trailer, which allows the tracking system to show what percentage of the trailer is unloaded. This data is available per teammate to see each individual's productivity.

Then, as trucks are re-packed by workers, the trailer is broken down into four cubes to ensure that cubic feet and weight are aligned during packing. This allows for greater efficiency both in packing and fuel required for the trailer (Anonymous, personal interview, 15 September 2025).

Based on all the available data, a significant portion of what would be needed for this exploration is already available. An initial study should be conducted to review all currently available data and assess processes within the RDC. The area that is believed to be missing data or requiring a study is the assignment of a truck to a door when it arrives at an RDC. This requires The Home Depot to move further back into their supply chain and obtain data from their suppliers to have more notice on what products are loaded onto which trucks and the expected arrival at which RDC. This would avoid the scramble when a shipment arrives without the RDC knowing its projected arrival time, and then needing to understand the contents of the trailer and find a dock that can receive it.

Data Preparation

All available data must be gathered and thoroughly explored after an initial understanding of the data is established. The first focus will be on collating all necessary data for the study and identifying any missing data that will be required to facilitate optimization at the RDC. This can be further broken down into detailing the necessary data sources and validating the quality of the data, and then identifying any potential issues.

Data Sources Needed:

Data Source	Data Type	Data Type	Data Type	Data Type	Possible Issue
Current The Home Depot RDC Data	Trailer data	SKUs on trailer	Product Type	Worker Data	THD data so these should present no issues
Supplier Shipment Information	Accurate shipment arrival dates; tracking trailer from leaving supplier center in route to THD RDC	Contents of trailer; known ahead of time			Would require access to supplier data; depending on the supplier could be an issue to obtain
Weather tracking	Public weather data				Public sources so no issues
Traffic Pattern	Traffic patterns				Public sources So no issues
FLOW Data	Purchase orders	Domestic warehousing & distribution centers	Domestic trucking	Domestic retailers & manufacturing	FLOW participants are able to see this data anonymized

Data from The Home Depot is available for use and will only require database access. Supplier shipment information could be more difficult to get from each supplier to see their current shipping habits, known delays, and driving schedules. The FLOW data helps participants forecast how current capacity and throughput will fare against future demand and enables anticipation of changes in the supply chain throughput and takes proactive steps to mitigate delays and fees. This data is anonymized and does not provide direct insight into specific participants.

Data Assumptions and Limitations

There are three key assumptions this study makes in the availability of data. These are: data reliability, stable infrastructure, and supplier compliance. For data reliability this study will

assume that historical data has been collected including shipment manifests, timestamps, and real-time sensor data. There are also assumptions that the data has been properly recorded, although some of these concerns will be addressed through the exploration and examination of data process. A stable infrastructure of the RDC, physical layout and total capacity (number of docks, available equipment) will remain constant throughout the initial modeling phase. It will also be assumed that suppliers, once segmented, will generally adhere to expected behavior patterns, allowing for predictive modeling to hold true.

Exploration and Examination of Data

RDC data can be studied to identify trends in suppliers' shipments, and visualizations can be created to see how often specific suppliers deliver products on time or early. This could also be cross-referenced with weather data to determine if weather was a factor. Potential data issues that could be identified now include missing values in the SKUs on a trailer field, which will be imputed using historical averages for that supplier, or flagged for manual review. All time/date fields must be converted to a uniform UTC format to ensure consistency across RDC systems and external weather/traffic feeds.

Since the problem being studied is highly dependent on timing (trailer arrival, unloading, dwell time), accurate time data is paramount. For this, all timestamps gathered from the disparate systems (RDC, FLOW, supplier data) must be converted to a single, consistent time zone, such as UTC, to ensure the proper calculation of time intervals. There may then be the need to create critical new variables by calculating the difference between existing timestamps. These could be for variables such as:

I. queue time: (time at dock door) - (time of trailer arrival at yard)

II. unloading efficiency: (time unloaded end) - (time unloaded start)

III. Total RDC dwell time: (time of store trailer departure) - (time of vendor trailer arrival)

The data could then be transformed into cyclical features, allowing the model to capture trends. This could be a day of the week to capture weekly supply chain rhythms, as well as a time of day. If the arrival time is transformed into a continuous cyclical variable (using a sine/cosine transformation), the model would then be able to understand that 11:59 pm is close to 12:01 am.

Missing data is expected, particularly from external suppliers and in manual logging fields. This could be captured in new variables (to help identify issues) such as:

Data Field/ Issue	Imputation Method	Additional notes/ Justification
Missing SKUs on Trailer	Historical mean for that supplier, flag for manual review	Provides a rapid, relevant estimate; manual flag ensures high-stakes errors are caught
Supplier Notice Time (Missing Tracking)	Conditional Mean Imputation (Historical mean specific to that RDC/supplier)	Acknowledges that notice time is dependent on the supplier's routine, providing a more accurate fill than a global average
Estimated Unload Time (Missing pre-arrival information)	Regression Imputation (Predict based on SKUs, cartons, product type)	Superior to simple mean; uses related, known variables to create a statistically robust estimate of required labor time
New Worker Productivity Metrics	Overall mean of all worker productivity metrics	Establishes a temporary, unbiased baseline until the new worker generates sufficient data points
Missing Single Shift Productivity	Last Observation Carried Forward (LOCF)	Assumes short-term performance continuity, which is often valid for established workers missing one scan/shift

I. The final step in the exploration and examination of data involves cleaning the data. Since logistics data is prone to human error, equipment failure logs, and network glitches, it could cause extreme outliers. For standardizing and data cleaning, there are a few steps that could be taken, including:

Data Cleaning/ Preprocessing Task	Description	Detail/Method	Target Variables / Data Type
Outlier Treatment	Identify and manage extreme outliers in calculated time variables	Use the Interquartile Range (IQR) method to identify outliers. Values should be capped at an upper threshold (e.g., a reasonable maximum) or entirely revoked (e.g., a 48-hour 'unloading time' is likely a logging error).	Time metrics • queue time • Unloading efficiency
Categorical Cleaning	Standardize text in free-text or manually entered fields	• Convert all text to lowercase • remove leading/ trailing space • use fuzzy string matching to correct common typos and map variations • e.g. ‘forklift’ vs ‘fork lift’ vs ‘forklif’ to a single, consistent category	Free-text or manually entered fields • Product Type • Dock Name • Dock Number
Normalization and Scaling	Transform values to ensure features measured on different scales contribute proportionally to a machine learning model	Use min-max scaling (normalization) or Z-score standardization	Numerical metrics • Dollar costs • Time in minutes • Count of SKUs

Modeling

Several algorithms or models could help determine the best course of action to suggest ways to improve The Home Depot's entire upstream supply chain process. The primary goal is to optimize resource allocation, including docks, personnel, and equipment, as well as routing within the RDC and outbound trailer packing. Three different categories of models can be used to analyze these issues and suggest optimizations: descriptive models, predictive models, and prescriptive models.

Descriptive models are helpful during the Exploration and Examination of Data phase, enabling the beginning of understanding patterns in the data by analyzing patterns and segmenting entities. These will assist in understanding the underlying patterns in the data, such as supplier behavior and worker performance variability. The models used during this phase would be K-Means Clustering and Association Rule Mining, also referred to as Apriori.

K-Means clustering groups data into clusters based on inherent similarities. This model is used to uncover hidden patterns or structures in the data. In this instance, suppliers would be clustered based on key behaviors, such as delivery punctuality, average "notice time" provided for shipments, and consistency of load type. The outputs of this analysis will help to inform different operational rules for each segment as well as identify outliers in the data. Visualizations, such as scatter plots and box plots, will also make outliers in the data clear to see. Outliers could range from errors, natural variations of the data, or infrequent events. Outliers must be reviewed to determine whether to remove, correct, or analyze them further.

To discover which products are consistently shipped together or which items frequently appear in the same store's allocation area, an Association Rule Mining (ARM) model, such as Apriori, can be used. ARM is used to identify the frequency of patterns in associations in item

sets to predict the next relevant item in the set. "This ARM technique is mostly used in business decisions according to customer purchases" (Nadeem, 2020). This is understood as the 'if-then' rule. The Apriori model is an algorithm that can be utilized to perform association rule mining, identifying frequently occurring items and relationships between them in the data. This will help generate association rules about relationships between items, such as "if item A is purchased, then item B is also likely to be purchased." According to Misti (2025), ARM relies on three key metrics: Support, the frequency with which an item appears in the dataset, Confidence, the likelihood that item B is purchased when item A is purchased, and Lift, the strength of a rule, measuring how much more likely item B is bought when item A is bought compared to when bought independently. This will help streamline the internal RDC flow and improve the efficiency of packing store trailers.

Predictive models will help forecast needs and are necessary to feed accurate information into the prescriptive models that will be utilized later in the process. Three models or techniques can be utilized during this phase, including Time Series Models, Regression Models, and Classification Models. The goal of these models is to forecast key timing and resource requirements, such as arrival time, unload time, and labor demand.

Time Series Models will "look at the patterns (or changes over time) within [the] data to understand it better and make predictions." (Korstanje, 2025.) There are several models that can be used for analyzing and forecasting data collected over time, including Autoregressive Integrated Moving Average (ARIMA), Prophet, and Long Short-Term Memory (LSTM). These time series models will allow for forecasting the expected time of arrival of vendor trucks and predicting future labor demand trends.

ARIMA allows for analyzing and forecasting time series data to account for patterns, trends, and seasonality. It will identify complex patterns based on past observations with future needs. The ARIMA model consists of three parts: autoregressive, integrated, and moving average. "The autoregressive component builds a trend from past values in the AR framework for predictive models." (Saadeddin, 2025.) This part is very similar to a regression model where lags of the time series' own past data values are used as the regressors. The integrated part is the difference of the time series component, so "trends and seasonality are eliminated… This step is necessary because it helps the model fit the data and not the noise." (Saadeddin, 2025). The moving average part centers around the relationship between the observation and the residual error. This part allows for inference about possible data trends by examining the present observation in relation to past errors.

Prophet is a forecasting model developed by Facebook for forecasting time-series data. "It works well for daily observations that exhibit patterns on different time scales" (Ahmed, J, 2024). According to Facebook, Prophet is robust to outliers, missing data, and dramatic changes in time series data. (Facebook, 2023.) Prophet is beneficial for time series data that has strong seasonal effects and special events. This is useful in the case of The Home Depot to make sense of holiday and summer sales events or purchase surges due to weather-related events.

Long Short-Term Memory (LSTM) is a deep learning Recurrent Neural Network (RNN) designed to model sequential data and learn long-term dependencies. This model will process time series data and predict the next future value in the series. (GeeksForGeeks, 2025.). LSTM has four components: the memory cell, which carries the data through the time steps; the forget gate, which determines what information to discard; the input gate, which determines what new

data to add; and the output gate, which controls what part of the state to output. This specialized gating "allows LSTM networks to selectively remember or forget information and [makes] them capable of capturing long-range temporal dependencies." (GeekForGeeks, 2025.)

The goal of utilizing these different Time Series Models is to predict arrival times by forecasting the accurate expected time of arrival of vendor trucks. These models use historical data, supplier shipment information, real-time tracking data, weather data, and traffic pattern data. All of this will provide the RDC with a reliable notice time to prepare the necessary dock and resources for unloading trailers. The Prophet model will be the primary model used due to its robustness to outliers, missing data, and ability to handle strong seasonal effects. ARIMA will be utilized as a performance benchmark.

Regression models, such as Multiple Linear Regression (MLR) or Gradient Boosting Machines, are used to predict a continuous target variable based on one or more independent variables and can be employed to estimate the unload time for a trailer. These regression models will utilize features such as the number of SKUs, the number of cartons/pallets, product type, supplier history, and the required unloading method (either manual or automated). These models and outputs are crucial for optimizing dock assignment.

Since there are multiple variables affecting the times for trailer unload, and not a single variable, the multiple regression model is the optimal choice for this study. "Multiple regression considers the effect of more than one explanatory variable on some outcome of interest [and] evaluates the relative effect of these explanatory, or independent, variables on the dependent variable when holding all the other variables in the model constant." (Hayes, A. 2025.) This allows for a clear view of what is driving changes and the strength of each factor's impact.

Gradient Boosting Machines (GBMs) begin with a simple tree model that provides an initial estimate of trailer unloading needs. A second tree is then trained, but not for predicting the variables, but rather for predicting the errors made by the first tree. The goal is to learn the pattern in the mistakes. A new prediction is then created by adding a small portion of the second tree's prediction to the first tree, and then another tree is trained to predict new errors. This process is repeated multiple times as each new tree 'boosts' the model's performance by focusing on what the model is getting wrong. "The name 'Gradient' comes from the underlying optimization technique. The process of fitting a new tree to the 'residuals' is a clever way of performing gradient descent on a loss function." (Kumar, D. 2025.)

Regression models will be used to estimate the continuous target variable, Trailer Unload Time. Due to the multiple features, SKUs, cartons/pallets, product types, required unloading methods, and supplier histories, a multivariate approach is necessitated. Gradient Boosting Machines will be deployed as they excel at minimizing errors sequentially and capture non-linear relationships better than standard Multiple Linear Regression models.

The next phase of modeling is prescriptive modeling. These models will determine the best course of action and suggest optimization and decisions to improve the RDC system. The focus is on optimizing resource allocation for docks, personnel, and equipment, as well as routing within the RDC and outbound trailer packing. The models and techniques employed during these phases include Linear Programming (LP)/Mixed-Integer Programming (MIP), Queuing Theory/Simulation, and Reinforcement Learning (RL).

Linear Programming (LP) and Mixed-Integer Programming (MIP) are optimization techniques commonly used by industries for supply chain optimization. Outputs from the

predictive models, such as predicted unload time and the trailer's contents (product type and size), required unloading method (automated, robot, manual), and current capacity and availability of suitable dock doors and required personnel/equipment, will all be used in LP and MIP models. The focus is on minimizing queue time, or the time it takes from a trailer's arrival to the assignment of a dock door, and maximizing throughput by finding the optimal, real-time assignment of incoming trailers to specific dock doors. A MIP Solver applies a branch-and-bound algorithm to divide the problem into smaller subproblems, which are then recursively solved to find the best solution. Each node of the branching tree solves a linear programming (LP) problem and provides either a lower or upper bound, and then compares the solution with the incumbent and decides whether to remove the branch (if it does not improve the current best solution) or branch further (to look for a new solution). "MIP can be used to optimize logistics, inventory levels, production schedules, and more." (GUROBI Optimization, ND.)

MIP is used for the real-time optimization of trailer-to-dock door assignments. MIP allows for the simultaneous optimization of multiple objectives (minimizing queue time and maximizing throughput) while still being subject to hard constraints (dock size, equipment availability, personal schedules). This technique is the industry standard for supply chain resource optimization and is a no-brainer for implementation within the RDC.

To help identify bottlenecks within the current upstream supply chain, Queuing Theory/ Simulation can be used to simulate the flow of trucks through the RDC, from yard arrival to unloading, to store departure, under various scenarios. "Queueing theory is a branch of mathematics used to describe, analyze, and predict the length of queues and waiting time in the system." (GeeksForGeeks, 2024.) Multiple models are created during this process to model the

arrival and service process, as well as the queue discipline of the model. The models are used with the data to reflect how the queue for trailers works, and then use probability theory and stochastic processes to deal with the randomness of real-world events (like trailer arrivals and unload/load times). The stochastic process models how the trailer process changes over time in random ways. After bottlenecks are identified, changes to the RDC system can be implemented, such as adding more docks or personnel.

"Reinforcement learning is a dynamic and adaptive process within machine learning algorithms that focuses on how agents, such as a computer program or software, for example, ought to take action in a specific environment to gain a reward." (Pal, 2024.) In this process, an environment model can be used to enable the agent, the RDC management system, to predict the consequences of its actions without interacting with the environment itself. This provides a controlled manner in which the agent can explore various strategies while simulating potential future states and rewards. Exploring the environment enables the agent to discover more effective strategies. Specifically, for this case, the focus of the RDC is on dynamic routing and inventory placement. The agent will learn the optimal internal movement strategies, including which path to take and where to temporarily stage goods, to minimize the total dwell time for products. It will receive rewards for efficiency and penalties for delays. This provides an adaptive, real-time optimization capability that traditional models lack.

Evaluation

Evaluating model performance is crucial for ensuring the suggested optimizations are accurate and reliable. Different metrics are required for each of the above detailed modeling phases.

Descriptive Model Evaluation

For K-Means clustering, the internal validation metrics ensure that discovered segments are distinct and meaningful. "Silhouette score is used to evaluate the quality of clusters created using clustering algorithms such as K-Means in terms of how well samples are clustered with other samples that are similar to each other." (Kumar, 2020). The silhouette score is calculated for every cluster, with a high score (closer to +1) indicating that clusters are dense and well-separated. A lower silhouette score (closer to -1) indicates that points may have been assigned to the wrong cluster. The Davies-Bouldin Index (DBI) is most commonly used to evaluate the goodness of a split by a K-Means clustering algorithm for a given number of clusters." (Sv, 2021.) DBI measures the average similarity ratio between clusters. The lower the score or average similarity, the better the clusters are separated, which indicates that clustering was performed well.

Predictive Model Evaluation

For forecasting and regression tasks, the quality of the prediction error will be measured. There are three measurements captured during this phase: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). RMSE is used to measure the standard deviation of residuals, or prediction errors. This gives a higher weight to

large errors and helps penalize significant timing mistakes. MAE measures the average magnitude of the errors. This measurement is easy to interpret as it is in the same unit as the target variable, in this case, minutes. MAPE is used for expressing prediction accuracy as a percentage. This makes it easy to compare against different targets/variables, such as expected time of arrival and unload time.

Prescriptive Model Evaluation

These models are evaluated based on the performance of the overall RDC system when the model's recommendations are applied. The measures captured during this phase will be the objective function value, average trailer dwell time reduction, dock utilization rate, and throughput. The primary metric for the MIP model is the objective function value. For the RDC, this involves minimizing total dwell time while maximizing dock utilization. Average trailer dwell time reduction will be a key business metric. This will be the measure of the average time a trailer spends in the RDC, from the time it enters the yard to the time it departs. Dock utilization rate is the percentage of time a dock door is actively serving a trailer, compared to its total available time. Throughput is a measurement of units per hour, measuring the volume of product that is moved through the RDC. This will validate the efficiency gains from the reinforcement learning-driven internal routing.

Limitations of the Approach

There are several limitations that need to be identified at the outset, including data sparsity and quality, external disruptions, and model latency. Poor or incomplete data for certain

suppliers or rare SKU types could severely limit the accuracy of the descriptive and predictive models that will be used. The models have limited capacity to predict or react to large-scale, unforeseen external disruptions and events such as port strikes, severe weather phenomena, or global pandemics. These events can all drastically alter supply chain dynamics and cannot be accurately predicted or addressed via models. Real-time optimization via MIP and RL requires low-latency data feeds and computational speed. If the RDC system cannot process new information and generate an optimal solution within seconds, its value for dynamic scheduling is diminished.

Ethical Considerations

As this venture directly involves monitoring worker activity and optimizing complex, high-stakes logistics, it would be remiss not to consider the ethical implications of this approach. There must be an ethical discussion to address the risk of creating a surveillance culture and using metrics to unfairly punish or stress workers, rather than identifying bottlenecks in the system. A mitigation of bias should also be discussed to ensure the productivity model accounts for external, non-performance factors, such as the time it takes to walk from a trailer to a specific area, equipment downtime, or the difficulty of a specific load (possibly bulk items). The focus of the optimization effort is to improve system efficiency, rather than ranking and penalizing individual workers.

Since the models' outputs focus on optimizing human and supplier resources, several ethical considerations must be proactively managed. Treating workers fairly and preventing burnout are top considerations from the outputs of the prescriptive models for optimizing

personnel allocation. This undertaking must be mindful of the risks of creating overly demanding or unstable schedules for RDC employees, which could lead to burnout or job dissatisfaction. The K-Means clustering results could create a 'tiering' of suppliers through the clustering process. Any operational rules created that are based on the segments must be transparent and justifiable to all suppliers and The Home Depot management. Any new rules must not unfairly disadvantage smaller suppliers or create negative contractual consequences. An overarching concern in any project like this is the privacy of the data. Any analysis of worker performance variability used for predictive labor demand must be aggregated and anonymized to protect individual employee privacy. Another top consideration is job displacement. The use of reinforcement learning to optimize routing and staging could eventually lead to the justification for increased automation within the RDC. This project must acknowledge and plan for the potential impact on human labor roles.

The Home Depot credits its success to being driven by its Core Values, one of which includes "Respect for All People and Taking Care of Our People." The Home Depot offers several programs to attract talent and upskill its employees, providing new career options within the Company to its associates. According to The Home Depot - Living Our Values Report (2024), "Our people strategy is rooted in the belief that empowered associates are the key to unlocking the full potential of our business." Given The Home Depot's already strong commitment to its associates, the ethical considerations of monitoring workers should not pose any concerns.

The Home Depot's values also include operating sustainably to reduce carbon emissions, promoting sustainable forestry, and conserving water, among other initiatives. The Home Depot

is committed to living its values, and this commitment is evident in the care it takes to support its people and the communities it serves. Each year, The Home Depot produces a "Living Our Values" report to highlight its associates and to provide new goals for operating sustainably and strengthening communities.

Deployment

As this is a hypothetical project study, no models will be deployed to a production environment. This paper serves as the final report for this effort, and a presentation will also be available for review.

If this were to be implemented in reality, the models would be deployed based on the collected data, and they would generate recommendations. The Home Depot would need collaboration from its suppliers to obtain the additional data it requires for optimizing trailers before they enter the RDC. This could pose a challenge if the suppliers have suboptimal tracking procedures or are unwilling to change the way they do business. Since THD has already made a strong effort with the Supply Chain Sync Initiative, automating processes and streamlining a significant portion of their upstream supply chain management, and has begun in-depth data collection, there is a strong belief that this initiative will be well poised for success.

Model recommendations would be utilized to optimize the RDC processes, such as improving the time for matching trailers to docks, trailer unload time, product movement around the RDC, and trailer load volume and times.

These optimizations would mean that products would be delivered faster and more efficiently to THD stores, and customers would have the supplies they need on store shelves,

ready for them. Customers would invariably be happier with the options in stock at their local THD and not have to go to other stores to purchase items. Happier customers mean more successful stores, which in turn will drive profits for the company. THD profits are shared with many of their staff, so staff wages would increase, as would employee satisfaction.

Conclusion

The Home Depot's (THD) upstream supply chain management, centered around the Rapid Deployment Center or RDC, will benefit from an in-depth study to suggest optimizations that can be done to speed up the time it takes to get a trailer into a dock, unload, move goods around the RDC, and then load a trailer for taking to a store. This study employed the CRISP-DM process to gain a comprehensive understanding of the business process and ensure that any created models addressed the core concerns of the upstream supply chain for the RDC. While THD has a large amount of data already covering this process, assumptions will be made based on the data, and it must be prepared and cleaned to ensure it is ready for use in the selected models. There are several categories of models for analyzing and suggesting optimizations, including descriptive models, predictive models, and prescriptive models. The desired outcome for the models is to optimize resource allocation regarding docks, personnel, and equipment for routing within the RDC and outbound trailer packing. While there are several ethical concerns, as this project centers on optimizing resources and incorporates both automation and a human element, if The Home Depot remains true to its core values, commitment to its staff, and belief that people are its greatest asset, the concerns will be mitigated.

References

Ahmed, J. (2024). **Time-Series Forecasting with TimescaleDB and Prophet.** TigerData. https://www.tigerdata.com/blog/time-series-forecasting-with-timescaledb-and-prophet.

Armstrong, C. (2011). **Excellence in execution at Home Depot.** *SupplyChainBrain*. https://www.supplychainbrain.com/articles/10855-excellence-in-execution-at-home-depot

Bond, J. (2015, February). **The Home Depot builds an omni-channel.** *Modern Materials Handling*. https://www.mmh.com/images/site/MMH1502.pdf

Deaton, J. (2023). **The Home Depot, Inc. 2023 investor conference.** *Seeking Alpha*. https://seekingalpha.com/article/4611314-the-home-depot-inc-hd-2023-investor-conference-transcript

GeeksForGeeks. (2024). **Queuing Theory.** *Geeks For Geeks*. https://www.geeksforgeeks.org/maths/queuing-theory/

GeeksForGeeks. (2025). **ARIMA vs Prophet vs LSTM.** *GeeksForGeeks*. https://www.geeksforgeeks.org/deep-learning/arima-vs-prophet-vs-lstm/

GUROBI Optimization. (n.d.). **MIP Solvers.** GUROBI Optimization. https://www.gurobi.com/faqs/mip-solvers/

Harvard Business School Alumni. (2019). **Using software tools to "remodel" Home Depot's Supply Chain.** Harvard Business School. https://d3.harvard.edu/platform-rctom/submission/using-software-tools-to-remodel-home-depots-supply-chain/

Hayes, A. (2025). **Multiple Linear Regression (MLR): Definition, Formula, and Example.** *Investopedia*. https://www.investopedia.com/terms/m/mlr.asp

Korstanje, J. (2025). **How to select a model for your time series prediction task [Guide].** *Neptune Blog*. https://neptune.ai/blog/select-model-for-time-series-prediction-task

Kumar, A. (2020). **Means Silhouette Score Explained With Python Example.** *DZone*. https://dzone.com/articles/kmeans-silhouette-score-explained-with-python-exam

Kumar, D. (2025). **Gradient Boosting Machines.** *Medium*. https://www.investopedia.com/terms/m/mlr.asp (This URL is a repeat and links to MLR, not GBM. Keeping the entry based on the title).

Mwidit, D. (2025). **Apriori algorithm explained: A step-by-step guide with Python implementation.** *Datacamp*. https://www.datacamp.com/tutorial/apriori-algorithm

Nadeem. (2020). **Association rules in machine learning and data-mining?** *Medium*. https://nadeemm.medium.com/association-rules-in-machine-learning-and-data-mining-f7e14eba8708

Pal, N. (2024). **What is reinforcement learning? An essential guide for tech professionals.** *EMERITUS*. https://emeritus.org/in/learn/what-is-reinforcement-learning/

Prophet. (n.d.). **Facebook.** https://facebook.github.io/prophet/

Ross, L. (2025). **The Home Depot's strategy for supply chain success.** *Thomas*. https://www.thomasnet.com/insights/the-home-depot-supply-chain/

Saadeddin, Z. (2025). **ARIMA for time series forecasting: A complete guide.** *Datacamp*. https://www.datacamp.com/tutorial/arima

Simply Wall St. (2025). **The Home Depot, Inc. (HD) leadership & management team analysis.** *Simply Wall Street*. https://simplywall.st/stocks/us/retail/nyse-hd/home-depot/management

SV, M. (2021). **Davies-Bouldin index for K-Means clustering evaluation in Python.** *Towards Data Science*. https://towardsdatascience.com/davies-bouldin-index-for-k-means-clustering-evaluation-in-python-57f66da15cd/

The Home Depot. (2018). **Supply chain elevates the Home Depot into Gartner Top 25.** *The Home Depot*. https://corporate.homedepot.com/news/company/supply-chain-elevates-home-depot-gartner-top-25

The Home Depot. (2024). **Living our values** [PDF]. https://corporate.homedepot.com/sites/default/files/2025-08/2025_Home Depot_Living_Our_Values.pdf

Trefis Team. (2018, August 16). **Home Depot delivers an impressive beat in the second quarter, driven by recovery in seasonal sales.** *Forbes*. https://www.forbes.com/sites/greatspeculations/2018/08/16/home-depot-delivers-an-impressive-beat-in-the-second-quarter-driven-by-recovery-in-seasonal-sales/#44fc621d316f

Trefis Team. (2018, June 13). **Why is Home Depot making a considerable investment in its supply chain?** *Forbes*. https://www.forbes.com/sites/greatspeculations/2018/06/13/why-is-home-depot-making-a-considerable-investment-in-its-supply-chain/

U.S. Department of Transportation. (2024). **Freight logistics optimization works.** https://www.transportation.gov/mission/office-secretary/office-policy/freight/flow/freight-logistics-optimization-works